Ray Zhao

rrz2016@nyu.edu | (612)-757-9612 | linkedin.com/in/rayruihengzhao | rayzhao.dev | github.com/rayz1459

EDUCATION

New York University **New York, NY**

B.A. Computer Science & Mathematics | Minor: Business Administration | GPA: 3.88 *Aug 2025 – May* 2028

Relevant Coursework: Data Structures, Computer Systems Organization, Calculus III, Probability and Statistics

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++

AI/ML: Transformers, CNNs, YOLO, NLP, TensorFlow, PyTorch, Hugging Face, Keras, NumPy, JAX, OpenCV

Other Domains: Signal Processing (EEG/EMG Brain waves), System Design/APIs, Microcontrollers, 3D printing, CAD

EXPERIENCE

NYU Stern Business Analytics Club (BAC) **New York, NY**

Machine Learning Engineer *Jan 2026 – Present*

- Selected as 1 of 16 participants from 100+ applicants for an ML mentorship, improving skills with hands on development
- Built multi-modal fusion model (DistilBERT + numeric MLP) and Random Forest for insider trading detection
- Built MLflow eval pipeline flagging 66% of insider events at 46× base rate precision across 44K held-out trades

Antaris.space **Fremont, CA**

Student Space Satellite Initiative *Oct 2023 – Jan 2025*

- Coordinated development of a 1U CubeSat across 3-person team, integrating CAD with software-hardware requirements
- Designed a standards-compliant 1U CubeSat satellite enclosure and mounting structures in Onshape
- Managed electrical, software, and mechanical interface specs, resolving subsystem conflicts across weekly design reviews

RESEARCH

NeuroLimbAI: AI Brain-Controlled Prosthetic Arm with Sensory Feedback *Nov 2023 – Jun 2025*

- Trained a Transformer model to translate 16-channel EEG signals into multi-DOF prosthetic arm movements with haptics
- Architected end-to-end brain-computer interface for multi-DOF arm at 95% online accuracy over 4 and 5 class sets
- Built a preprocessing pipeline to resample 256 Hz signals to 100-step sequences, apply PCA, and normalize amplitudes
- Implemented positional encoding, multihead self-attention, and layer normalization; won GENIUS Olympiad Silver Medal

VertiPaw: Wall-Climbing Quadruped for Infrastructure Inspection *Jan 2024 – Dec 2024*

- Developed suction based quadruped to climb vertical surfaces with finite-state gait controller attaining 9lb payload capacity
- Integrated LiDAR-based localization and orientation estimation enabling automated wall approach and climbing transitions
- Presented at IEEE Int'l Conf. on Robotic Computing (IRC) 2024, Tokyo, Japan

PROJECTS

Custom Convolutional Neural Network System with Manual Backpropagation in Java *Sep 2025 – Dec 2025*

- Built a CNN from scratch in Java with forward/backpropagation and convolutional layers, without external ML libraries
- Designed a modular OOP pipeline for image classification (cats vs. dogs) with He initialization, preprocessing, and L2
- Achieved 91% accuracy image classification on a balanced cats-vs-dogs dataset using the fully custom Java training pipeline

Natural-Language-Commanded Autonomous Microbot Swarm with Learned Stealth Policy *May 2026 – Jul 2026*

- Built a stealth microbot-swarm simulator where 10 agents cooperatively map environment while evading hunting observer
- Engineered LLM pipeline (Llama 3.1 function calling + schema validation) mapping text to executable swarm behaviors
- Implemented CEM policy-based RL optimizing exposure/coverage reward reducing detection by 10% vs greedy baseline

PUBLICATIONS

NeuroLimbAI: Enhancing Sensory Feedback in an Origami-Inspired Prosthetic Arm with EEG-Controlled Noninvasive Vibrotactile Haptic Feedback Published in IEEE ICAPAI 2024, ISBN: 979-8-3503-4977-1

VertiPaw: Development of a Vertical Climbing Robot with Adaptive Suction and Suspension Foot Design for Infrastructure Monitoring Published in IEEE IRC 2024, ISBN: 979-8-3315-2156-1